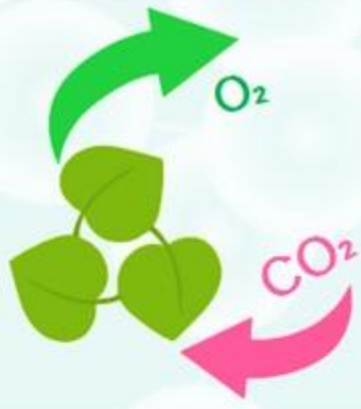




BIOLOGY



HEREDITY AND EVOLUTION

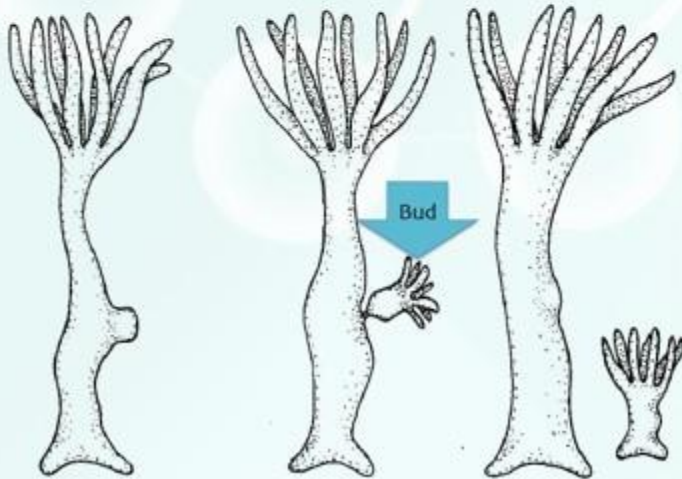
MODULE - 16



REPRODUCTION

ASEXUAL

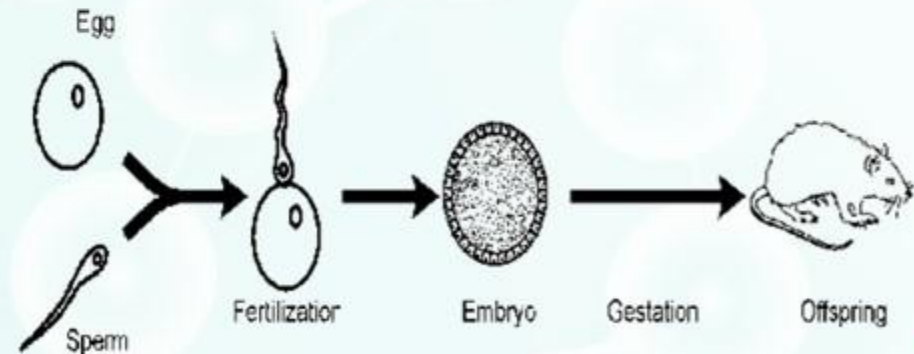
Offsprings have the same genetic makeup like the parents. These offsprings have no variation and they are exact clones of their parents



SEXUAL

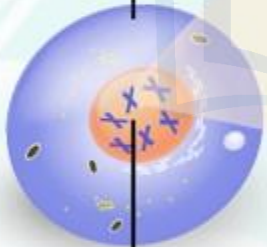
Offsprings have different genetic makeup due to recombination

Variation in the organisms



HEREDITY

Cell



Nucleus



Chromosome

DNA



Gene

Nucleus of cell



23 pairs of Chromosomes -
thread like structures



DNA molecules



A part of DNA responsible for
a special trait - Genes



Equal amount of genetic
material is contributed to
child from both the parents



EXPRESSION OF GENES

PHENOTYPE

Its how an organism looks
or it shows the physical
'traits'



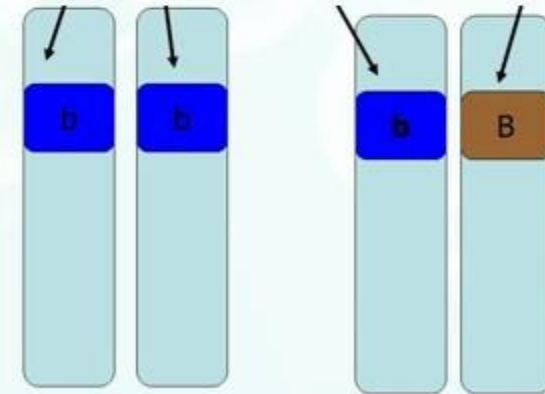
GENOTYPE

Its an organism's genetics
This gets expressed into the
'traits'



ALLELES

For each trait, there are two factors coming
from each parent. This is called alleles.
Alleles can be identical – **homozygous**
Alleles can be non-identical – **heterozygous**



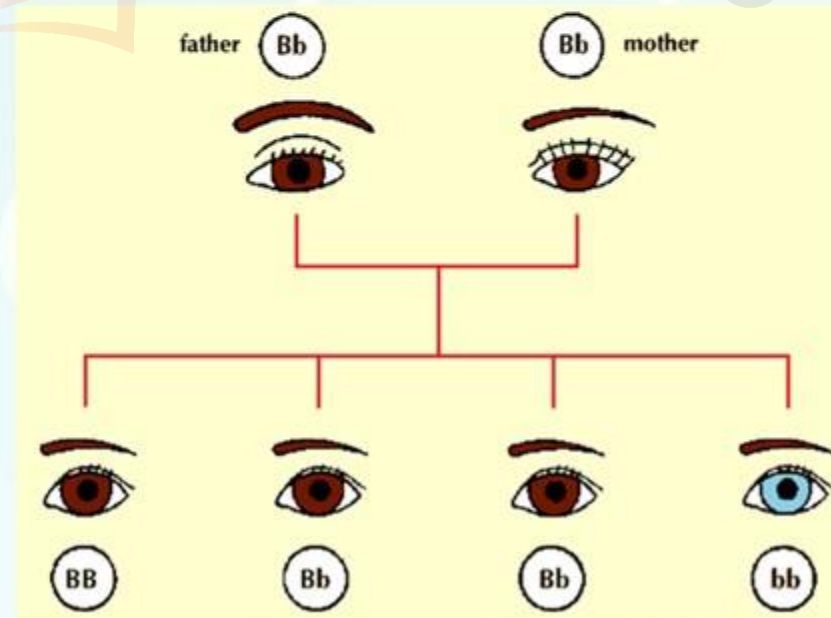
HETEROZYGOUS ALLELES

DOMINANT TRAIT

The trait which gets expressed over all other traits. It is genotypically expressed with a capital letter.

RECESSIVE TRAIT

The trait which is not expressed over other traits is a recessive trait. Dwarfism is represented by small letter.



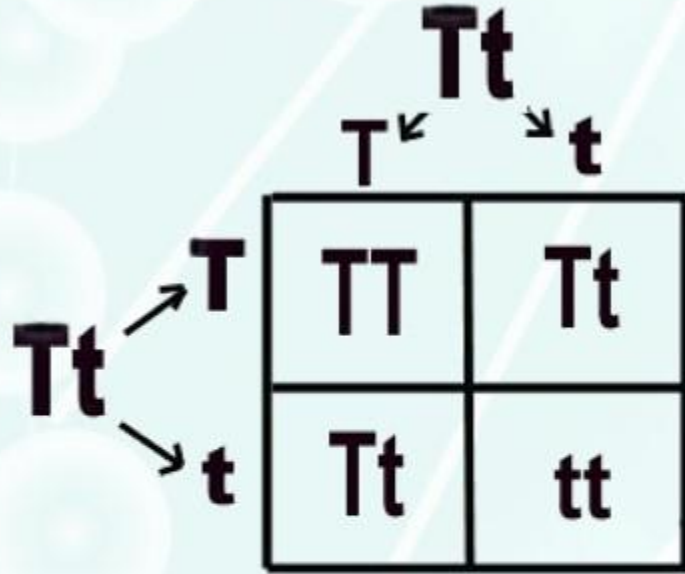
GREGOR J. MENDEL – THE FATHER OF MODERN GENETICS



- Mendel's pea plant experiments conducted between 1856 and 1863 established many of the rules of heredity, now referred to as the laws of Mendelian inheritance.
- Mendel worked with seven characteristics of pea plants: plant height, pod shape and color, seed shape and color, and flower position and color.
- Tschermak, Hugo de Vries, Correns, and Jasper Spillman independently verified several of Mendel's experimental findings, ushering in the modern age of genetics.



MENDEL'S LAWS OF INHERITANCE



1

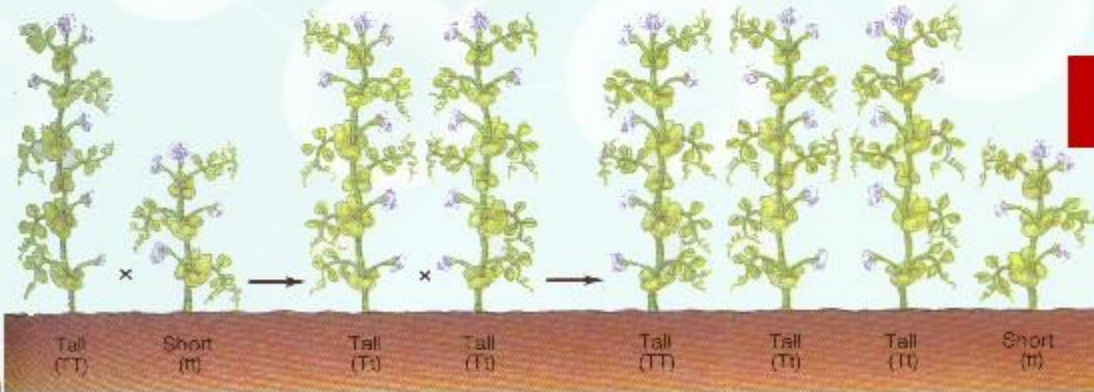
Law of segregation - During gamete formation, the alleles for each gene segregate from each other so that each gamete carries only one allele for each gene.

2

Law of independent assortment - Genes for one trait are not inherited together with another trait.

3

Law of dominance - Some alleles are dominant while others are recessive; an organism with at least one dominant allele will display the effect of the dominant allele.



DOMINANCE

Complete dominance

The allele that is regarded as dominant completely masks the effect of the recessive allele



Incomplete dominance

One allele does not completely dominate another allele, and therefore results in a new phenotype



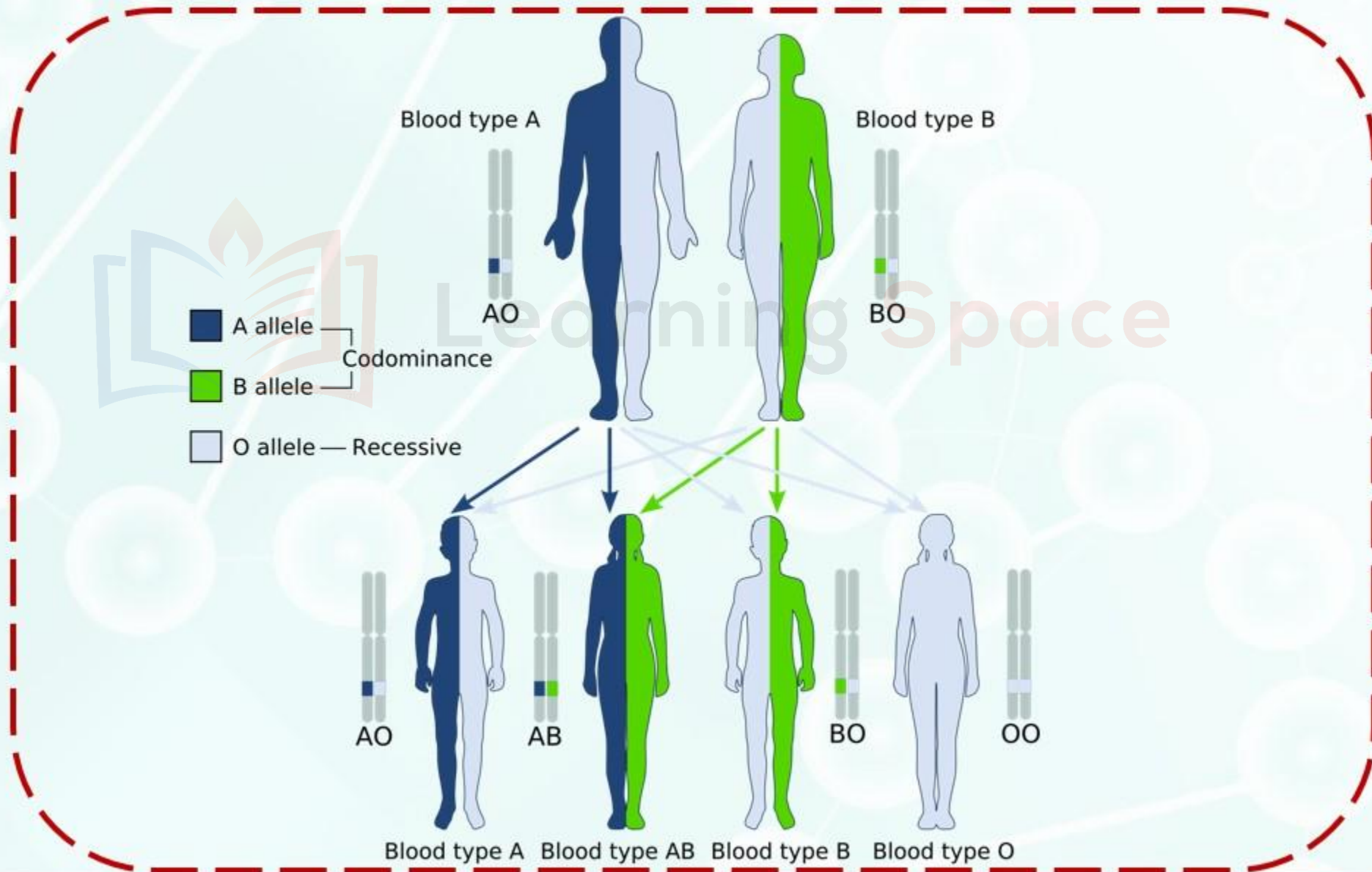
Co-dominance

The alleles of a gene pair in a heterozygote are fully expressed. The offspring is a phenotype that is neither dominant nor recessive

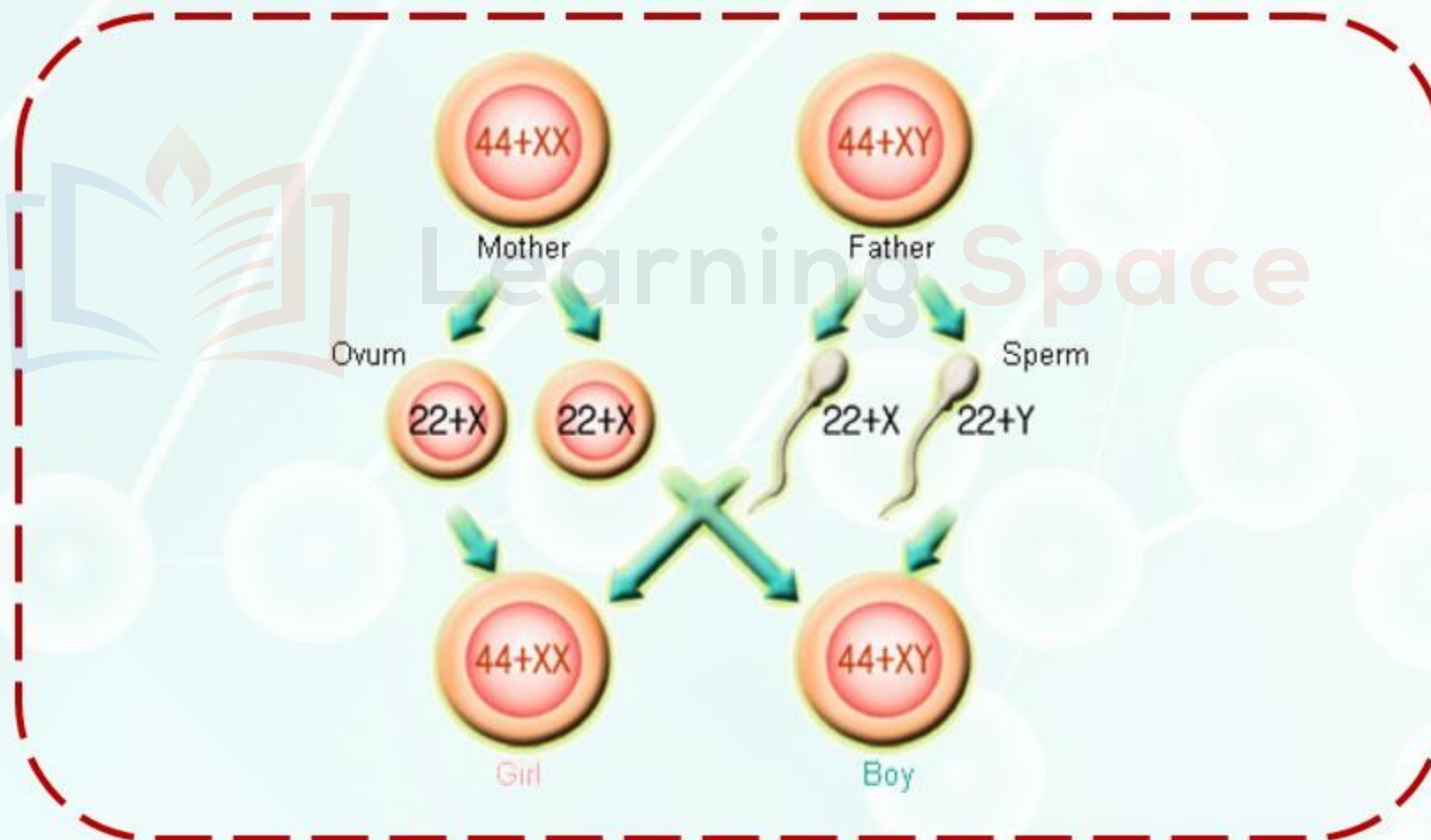
Blood type	Genotype	
A	I^A, I^O	AO
	I^A, I^A	AA
B	I^B, I^O	BO
	I^B, I^B	BB
AB	I^A, I^B	AB
O	I^O, I^O	OO



CO-DOMINANCE IN HUMAN BLOOD GROUP

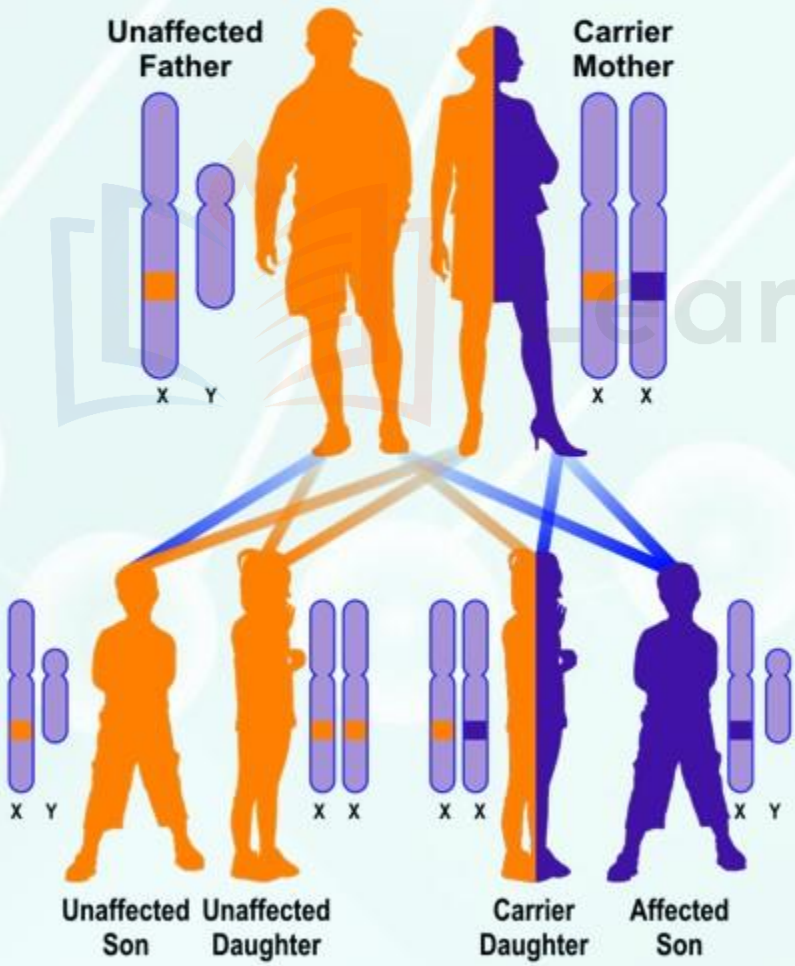


SEX DETERMINATION IN HUMAN BEINGS

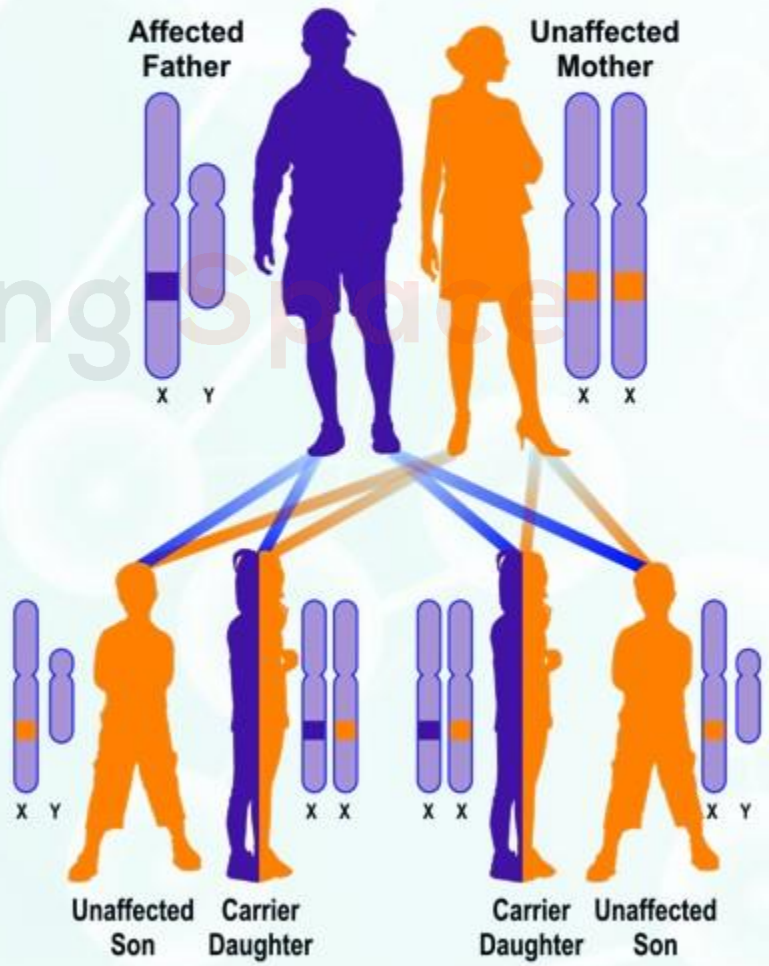


X-LINKED RECESSIVE INHERITANCE

X-linked Recessive, Carrier Mother



X-linked Recessive, Affected Father



X-LINKED RECESSIVE INHERITANCE

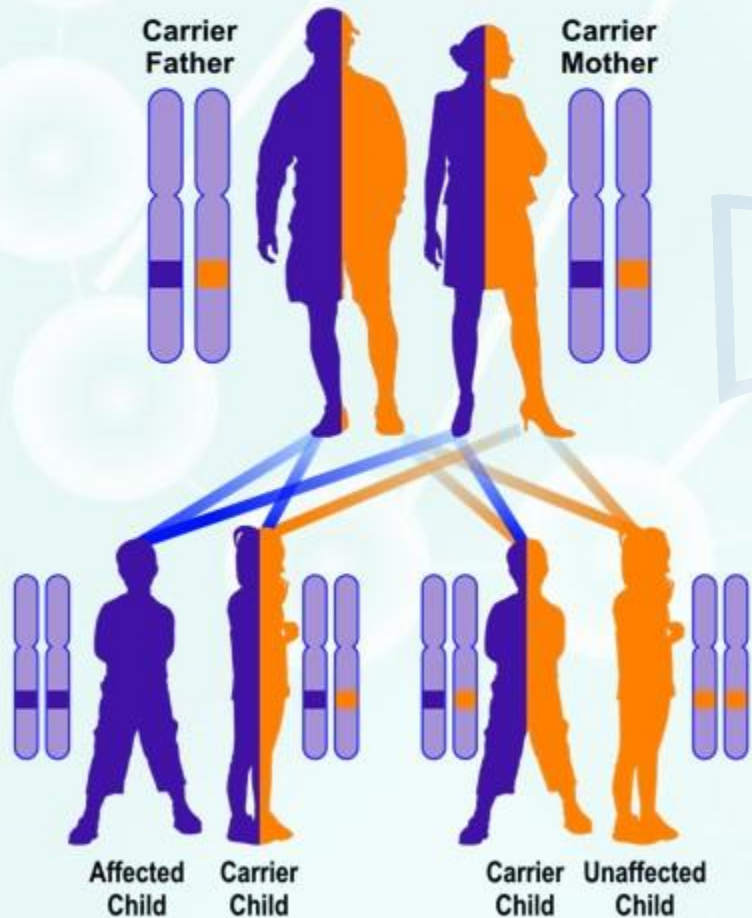


- **Alteration in the X chromosome causes the medical condition in males but not in females.**
- As males have only one X chromosome, receiving a carrier gene will affect them.
- In case of females, they receive both the X chromosome, thus they are prevented by the dominant X gene.
- Thus, females can be carriers whereas males are affected.
- The disease is passed from carrier mothers to their sons.
- Examples- colour blindness, hemophilia, muscular dystrophy



AUTOSOMAL RECESSIVE INHERITANCE

Autosomal Recessive



- Affects the offspring only when both parents are carriers of the disease.
- 25% chance that the offspring will be born with severe disease.
- Examples- Sickle cell anemia, Thalassaemia and Cystic fibrosis

Sickle cell is linked to chromosome number 11 and thus it can occur in both males and females.

RBC obtain a crescent sickle shape leading to anemia and can be fatal.



BASICS OF EVOLUTION

INHERITED TRAIT

- The changes are transferred to the forthcoming generations.
- They are genetic variations



ACQUIRED TRAIT

- The changes are not transferred to the forthcoming generations.
- They are somatic variations



EVOLUTION IS NOT PROGRESS!



- Evolution only brings complexity in organisms and not superiority.
- It is not necessary that old species will vanish after evolution.
- For e.g. chimpanzees and humans both have a common ancestor but this does not mean that humans evolved from humans.
- Evolution happens because of
 - Genetic variation
 - Natural selection.



EVOLUTIONARY RELATIONSHIPS



HOMOLOGOUS ORGANS – Wing of a bat, lateral fin of a whale and a human arm; perform different functions but have same internal structure



ANALOGOUS ORGANS - Butterfly's wing and bat's wing; both have same function but different in internal structure which are not inherited from a common ancestor.



FOSSILS

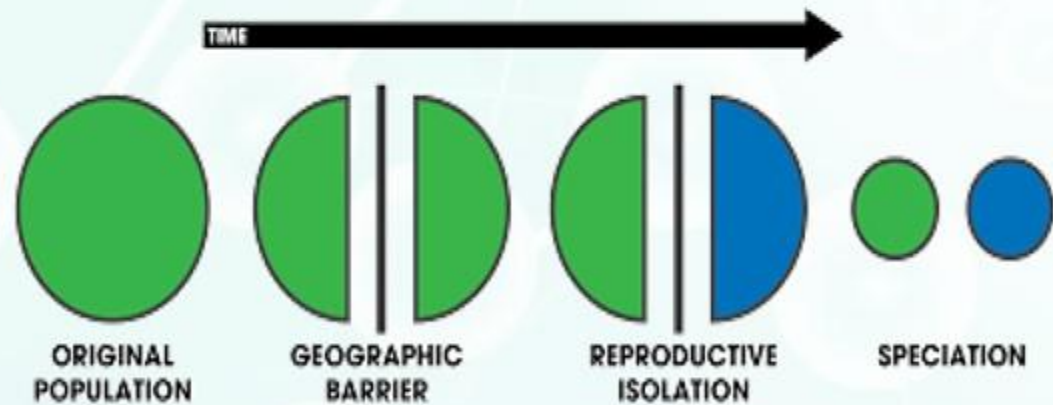
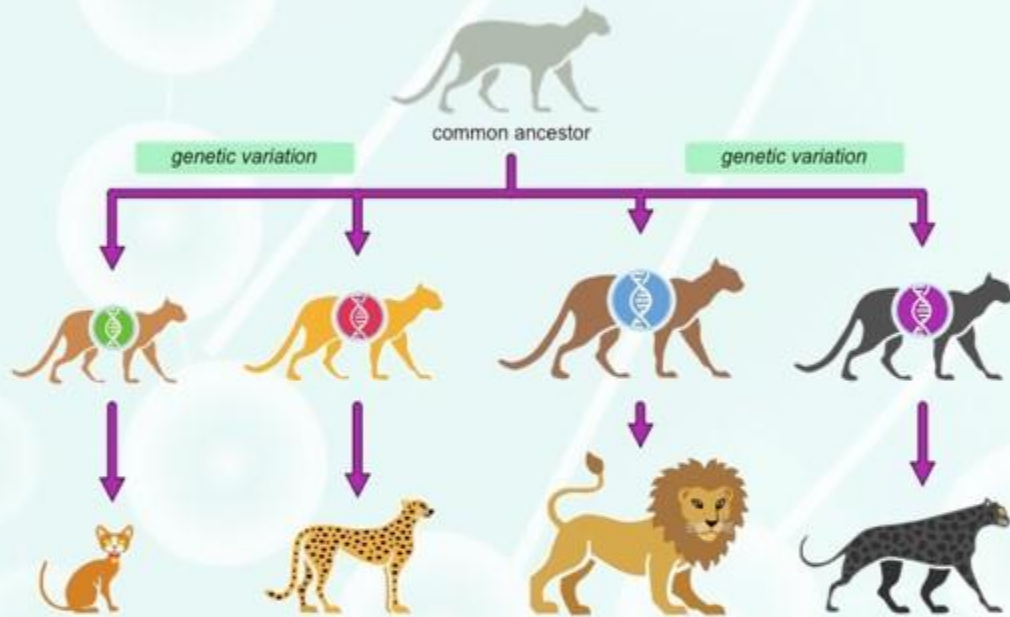


- Preserved remains or impressions of living organisms
- It helps us to understand the difference between current species and extinct species.
- Helps to establish common ancestry amongst different species.
- Informs about the evolution of different species on Earth.
- The age of the fossils can be calculated by
 - Radioactivity and carbon dating
 - Depth of fossils

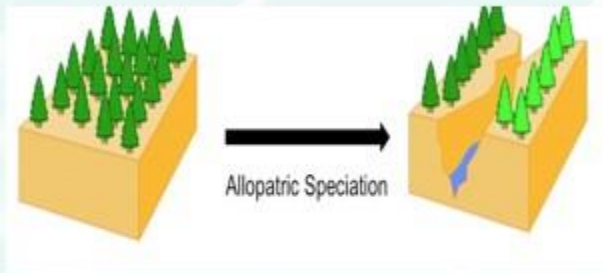


SPECIATION

- The process by which new species develop from the existing species.
- Speciation leads to formation of two or more species from an original population who cannot interbreed.

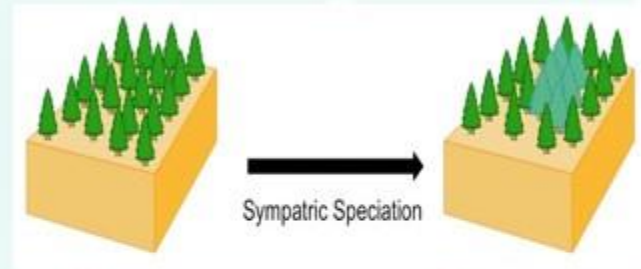


SPECIATION – TYPES OF CAUSES



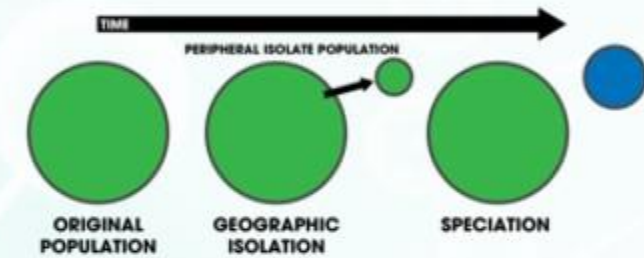
ALLOPATRIC SPECIATION

It occurs when the population of a species undergo reproductive isolation due to a physical barrier



SYMPATRIC SPECIATION

It occurs when two different species descend from the same ancestor. Both the species thrive at the same location



PERIPATRIC SPECIATION

When a small amount of population undergo reproductive isolation from the main population(genetic drift)



EVOLUTION OF HUMAN

The human lineage

*Australopithecus
afarensis*



Homo habilis



Homo erectus



Homo neanderthalensis



Homo sapiens



40 cm
12 inches



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